Surviving in the Era of Artificial Intelligence

In 2025, one of the last farmers in the Mexican state of Querétaro stood helplessly upon his once-vibrant, now-barren fields as his hometown became the "promised land" of artificial intelligence (AI) datacentres. Querétaro recently endured centuries' worst drought due to substantial water consumption by tech giants' warehouses full of computer servers, impacting crops and water supplies to nearby communities (Bearne, 2025).

With the tremendous surge in Generative AI's popularity, more datacentres are being built around the globe. However, this popularity comes with a huge cost. There are thousands of servers running in these datacentres, converting massive amount of electricity to waste heat. To prevent overheating, datacentres use cooling systems; mainly water-based (WBC) and air-based cooling (ABC), where the latter operates with air-conditioners and fans. Ironically, these systems consume billion litres of water annually and account for 37% of total electricity consumed by datacentres (Mytton, 2021; Chen et al., 2025).

This is evident from how Google and Microsoft, 2 dominant leaders in the AI industry, consumed 8 billion gallons of water in 2023 alone (Figure 1). Most datacentres, 42% of Microsoft's and 15% of Google's, are located in water-stressed areas like Virginia and Mexico (Hodgson, 2024). Consequently, citizens face a severe water crisis for basic needs and agriculture; raising conflicts of whether water is more important for fulfilling people's needs or to fuel technological industries.

As the 4th largest datacentre market, Hong Kong (HK) is rapidly scaling its datacentre development, which account for more than 6% of total energy consumption sourced largely from fossil fuels (TVB, 2026). Research discovered that datacentres' heavy dependence on fossil fuels for meeting energy demands will emit 2.5Bn tonnes of greenhouse gas emissions by 2030, an increase from 330M before the development of AI (Tiberi et al., 2025). Still, HK lacks a clear sustainability plan and transparent power usage effectiveness (PUE) requirements for datacentres. Although this attracts more datacentres, the resulting energy demand and carbon emissions hinder HK from achieving carbon neutrality by 2050.

Collectively, AI's hidden water and carbon footprint will compete with citizens for a finite resource globally and lower climate quality in HK.

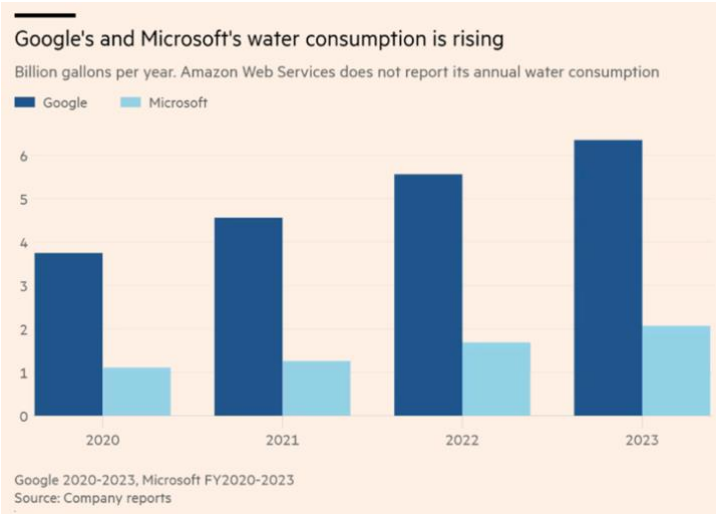

Figure 1: Google's and Microsoft's Datacentre Water Consumption 2020-2023

© Hodgson, C. (2024, August 18). US tech group's water consumption soars in 'data centre alley'. *Financial Times*. https://www.ft.com/content/1d468bd2-6712-4cdd-ac71-21e0ace2d048?syn-25a6b1a6=1

This paper suggests a cooling model, **liquid immersion cooling**, that improves water and energy usage simultaneously to support sustainability.

Immersion cooling involves submerging electronic hardware in a dielectric fluid, like mineral oils, that have better electrical neutralising properties and higher heat capacity than water (Kuncoro et al., 2019). This replaces water-intensive evaporative cooling in traditional cooling towers, eliminating the need for billion gallons of freshwater. Furthermore, research observed that datacentres utilising immersion cooling systems achieve a 60% reduction in energy consumption compared to traditional cooling methods (Liu et al., 2025). By alleviating the need for computer room air-conditioning and fans, immersion cooling can potentially reduce the PUE to 1.1 and save around 77.2% of CO_2 emissions compared to ABC (Figure 2; Figure 3).

Firstly, water-stressed regions' governments should mandate big companies like Microsoft and Google to retrofit all existing datacentres with liquid immersion cooling systems within a targeted timeframe. This ensures no family in Querétaro faces power outages and future generations receive high quality water.

Secondly, HK government should set stricter sustainability measures for datacentres, like a PUE limit of 1.3 and requirement for datacentres to replace ABC and WBC with immersion cooling. In 2024, HK University of Science and Technology (HKUST) implemented HK's one of the fewest immersion cooling technologies for sustainable AI research. Following the implementation, HKUST reported an 80% reduction in energy consumption for cooling and anticipated avoiding 900 tonnes of carbon emission annually, which is 45% less

than ABC (Siu, 2024). Hence, immersion cooling can benefit HK residents' life quality by reducing carbon emissions and help the government achieve 2050 neutrality goal efficiently.

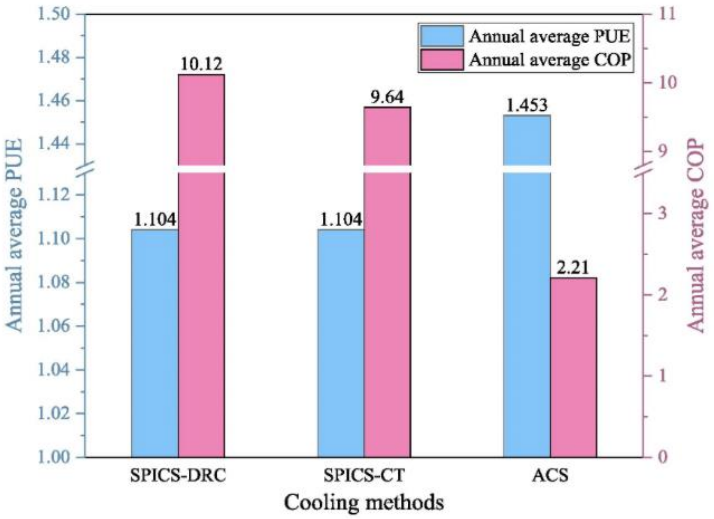

Figure 2: Comparison of PUE and COP for various single-phase immersion cooling (SPIC)

©Liu, S., Guo, S., Sun, H., Xu, Z., Li, X., & Wang, X. (2025). Evaluation of energy, economic, and pollution emission for single-phase immersion cooling data center with different economizers. *Applied Thermal Engineering*, *260*, 125049.

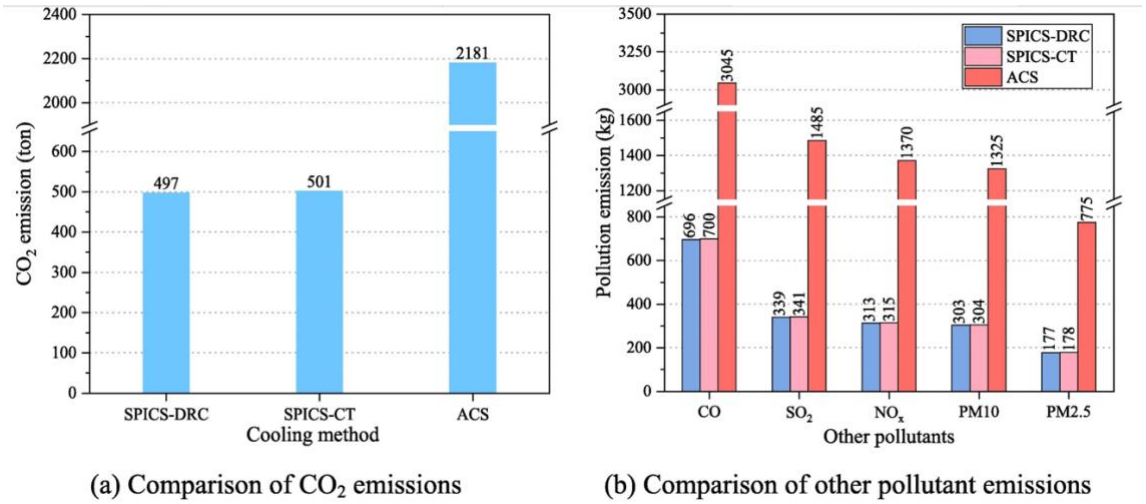

Figure 3: Comparative pollutant emissions (tons) of various datacentre cooling technologies: SPICS-DRC, SPICS-CT, and ACS

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Policy alone isn’t enough. we must take responsibility for our digital footprint.

GPT-3 version of OpenAI’s ChatGPT drinks approximately 500ml of freshwater for answering every 20 to 50 questions (Gendron, 2023). Hence, before using AI for trivial tasks or making *Ghibli* arts for a mere *Instagram* post, one must ask: “Is this worth a bottle of water?”

While Microsoft and Google’s water consumption is rising, Amazon Web series does not report its annual water footprint at all. Thus, instead of opting for free AI platforms, consumers should choose providers that publish water and energy statistics transparently.

Instead of waiting for others to take the initiative in healing the climate, let’s start improving our own consumption habits to make a greater impact immediately.

739 words

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